

MACQUARIE ISLAND, Australia

WMO: 949980

Lat: 54.499S

Lon: 158.937E

Elev: 8

StdP: 101.23

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.9	-1.7	-7.9	1.9	-1.9	-6.7	2.1	-0.9	21.3	4.7	20.0	4.6	8.6	190	1.217

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.6	9.7	8.6	9.2	8.2	8.9	7.9	9.1	9.4	8.7	8.9	8.2	8.6	10.1	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.1	9.2	8.4	6.9	8.8	8.0	6.7	8.4	27.5	9.4	26.5	9.0	25.7	8.7	12.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	18.3	16.8	DB	-4.7	10.9	1.9	0.8	-6.0	11.5	-7.1	12.0	-8.2	12.5	-9.5	13.1	(4)
(5)			WB	-5.8	10.0	1.6	0.6	-6.9	10.5	-7.9	10.8	-8.8	11.2	-9.9	11.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	6.9	7.1	6.4	5.6	4.5	3.6	3.2	3.5	3.6	4.0	4.6	6.1
	DBStd	2.12	1.12	1.13	1.22	1.41	1.82	2.06	2.09	1.75	1.83	1.51	1.57	1.39
	HDD10.0	1861	96	83	111	131	171	193	211	202	193	187	163	120
	HDD18.3	4903	354	316	369	381	429	443	469	461	443	446	413	378
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	10.2	9.5	10.0	10.5	10.9	10.5	10.3	10.3	10.1	10.3	10.6	9.8	9.1
Precipitation	PrecAvg	953	84	85	96	93	86	78	74	73	74	78	73	80
	PrecMax	1243	175	130	165	174	144	155	126	156	134	128	138	177
	PrecMin	717	30	24	30	50	50	30	22	27	25	38	34	34
	PrecStd	126	28	24	27	25	24	26	26	29	24	24	22	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	10.4	9.7	9.1	8.4	7.5	7.2	7.1	7.2	7.7	8.6	10.2
		MCWB	8.7	9.4	8.8	8.3	7.8	6.6	6.5	6.3	6.3	6.7	7.2	9.1
	2%	DB	9.6	9.7	9.1	8.6	7.7	7.0	6.7	6.6	6.7	7.1	7.7	9.3
		MCWB	8.3	8.7	8.4	7.9	7.0	6.2	6.0	5.9	5.9	6.1	6.6	8.3
	5%	DB	9.1	9.2	8.7	8.2	7.2	6.6	6.2	6.2	6.4	6.6	7.2	8.7
		MCWB	8.0	8.2	7.9	7.5	6.6	5.9	5.6	5.6	5.7	5.8	6.2	7.7
	10%	DB	8.7	8.7	8.3	7.7	6.8	6.2	5.8	5.9	6.1	6.2	6.7	8.1
		MCWB	7.6	7.8	7.5	7.1	6.2	5.7	5.3	5.3	5.4	5.4	5.8	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	9.9	9.3	8.7	8.1	7.0	6.7	6.7	6.7	7.2	7.7	9.7
		MCDB	9.8	10.2	9.4	8.8	8.2	7.1	6.9	6.8	6.9	7.4	8.1	9.9
	2%	WB	8.8	9.1	8.7	8.2	7.2	6.5	6.2	6.2	6.3	6.6	7.1	8.7
		MCDB	9.1	9.3	8.9	8.4	7.4	6.7	6.5	6.4	6.5	6.8	7.4	9.0
	5%	WB	8.3	8.6	8.2	7.7	6.8	6.1	5.8	5.8	6.0	6.1	6.6	8.0
		MCDB	8.7	8.9	8.5	8.0	7.1	6.4	6.1	6.1	6.3	6.5	7.0	8.4
	10%	WB	7.9	8.2	7.7	7.2	6.4	5.8	5.4	5.5	5.6	5.7	6.1	7.5
		MCDB	8.4	8.5	8.1	7.6	6.7	6.2	5.8	5.8	5.9	6.1	6.6	8.0
Mean Daily Temperature Range	5% DB	MDBR	2.8	2.6	2.6	2.5	2.7	2.8	2.7	2.7	2.9	3.1	2.9	2.9
		MCDBR	3.2	2.9	2.7	2.5	2.7	2.6	2.7	2.7	2.7	3.0	3.1	3.2
		MCWBR	2.9	2.8	3.1	2.9	3.3	3.0	3.1	3.2	3.0	3.2	2.9	2.8
	5% WB	MCDBR	2.9	2.6	2.7	2.5	2.6	2.6	2.7	2.7	2.7	2.9	2.8	2.9
		MCWBR	2.9	2.8	3.1	3.1	3.2	3.0	3.2	3.2	3.0	3.2	2.9	2.7
Clear-Sky Solar Irradiance	taub	0.333	0.331	0.329	0.319	0.315	0.298	0.299	0.321	0.338	0.333	0.329	0.328	
	taud	2.535	2.551	2.542	2.510	2.316	2.265	2.362	2.444	2.440	2.488	2.520	2.549	
	Ebn at Noon	967	930	864	758	591	519	609	735	834	916	965	982	
	Edh at Noon	101	91	80	64	54	45	50	66	87	96	102	101	
All-Sky Solar Radiation	RadAvg	5.01	3.95	2.51	1.30	0.62	0.37	0.50	1.04	2.10	3.44	4.63	5.23	
	RadStd	0.31	0.20	0.18	0.09	0.06	0.03	0.04	0.09	0.11	0.27	0.30	0.41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.25	+0.53	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page